

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

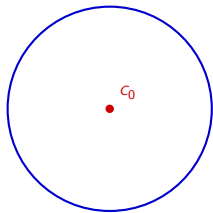
Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point
 AA' → Directing Line
Axis → Center Line

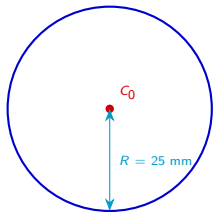


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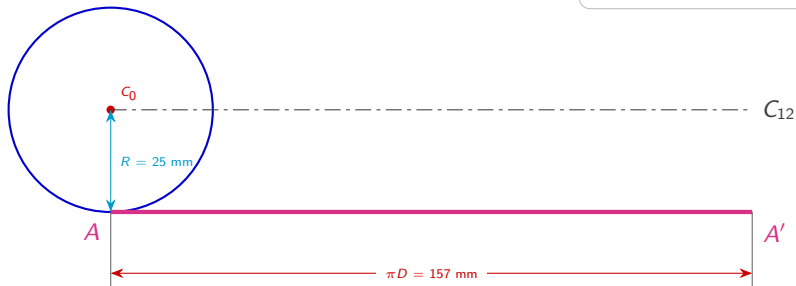
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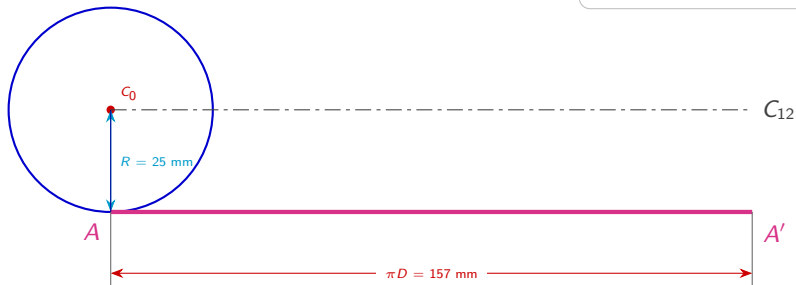
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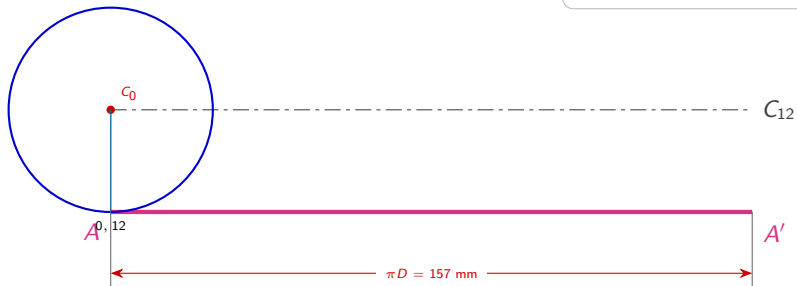
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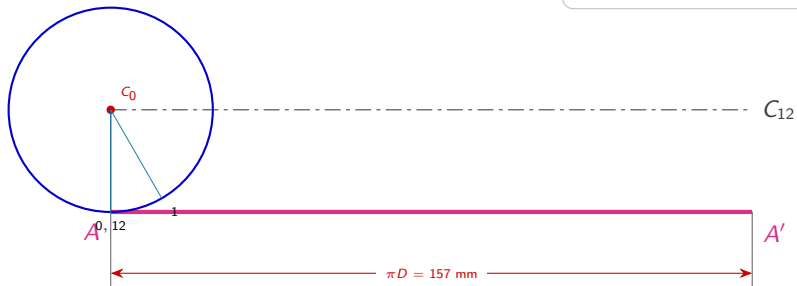
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



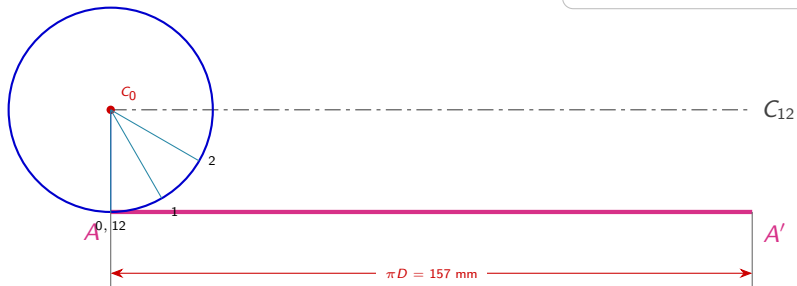
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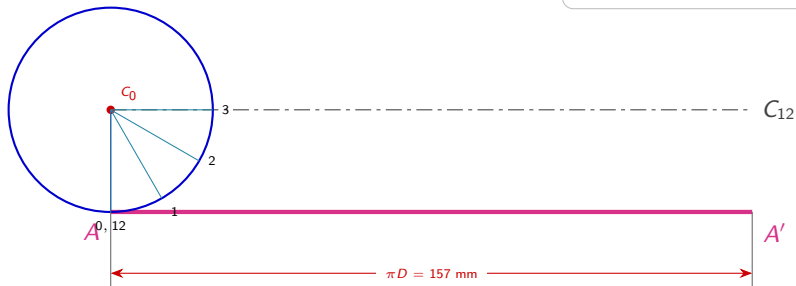
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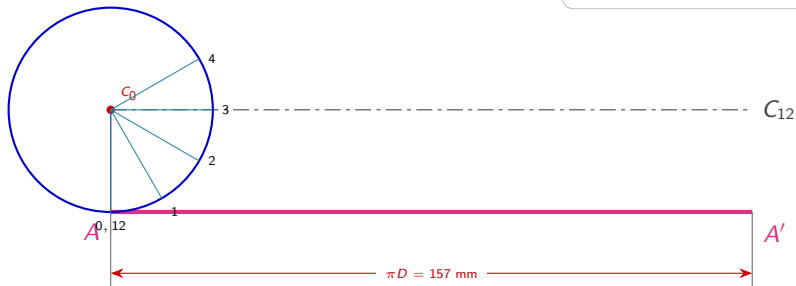
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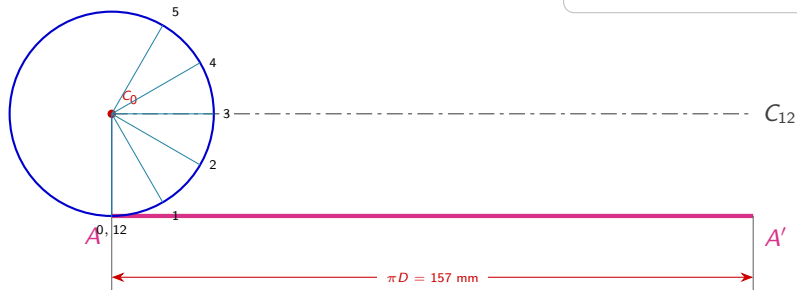


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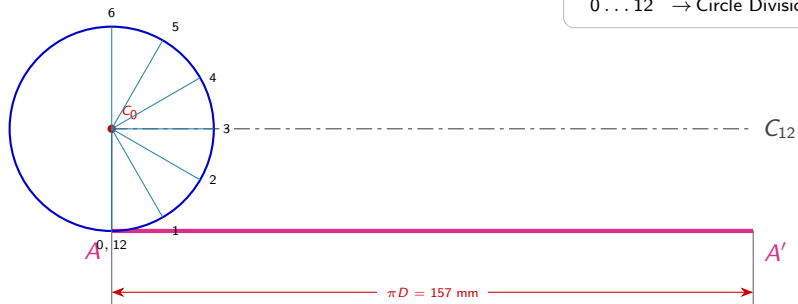
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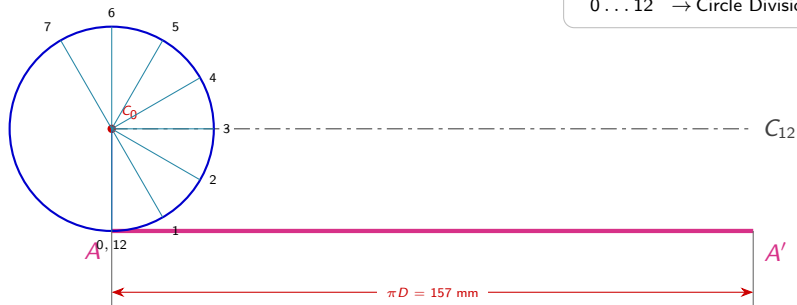
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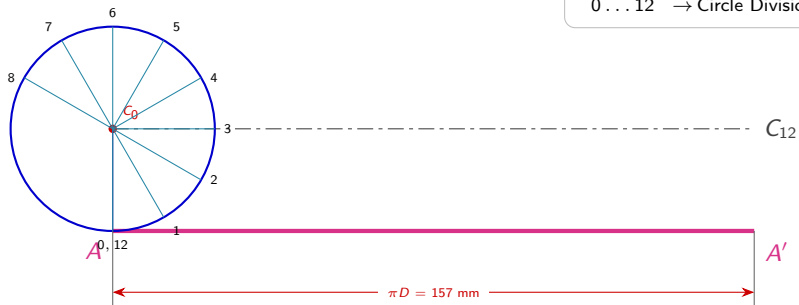
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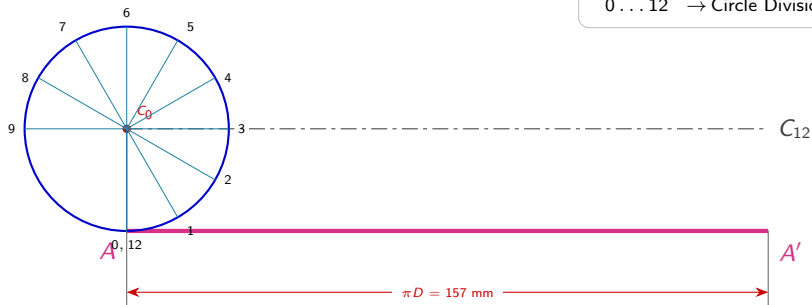
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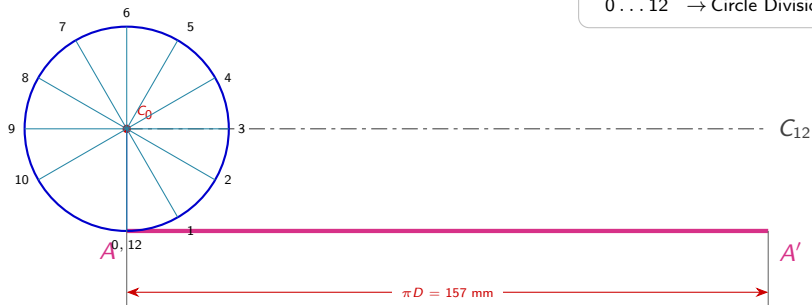
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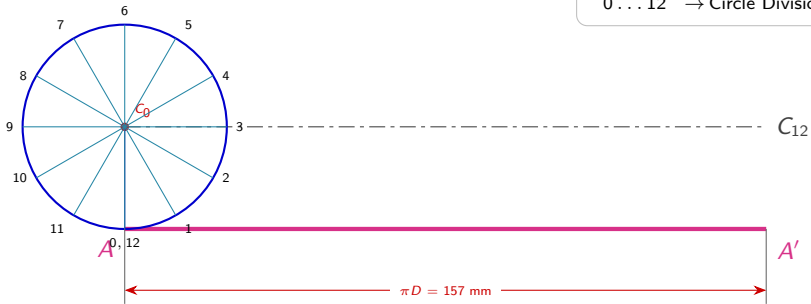
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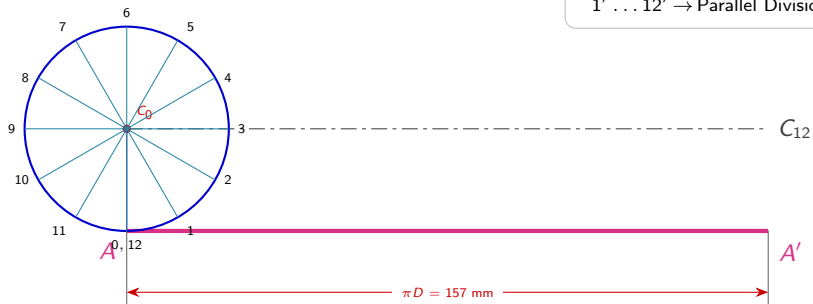
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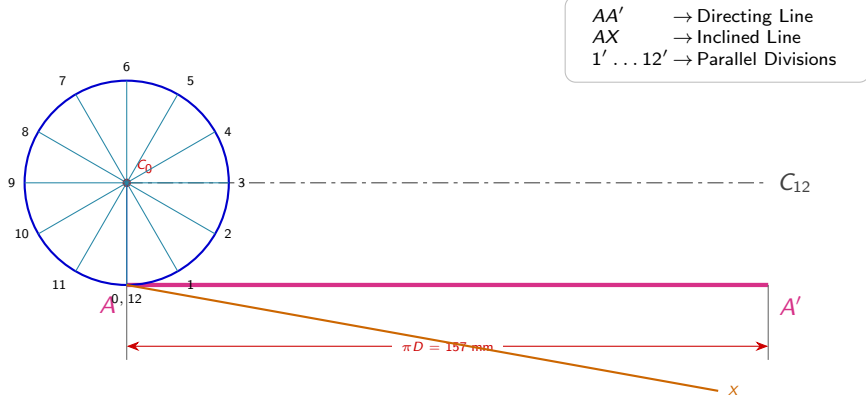
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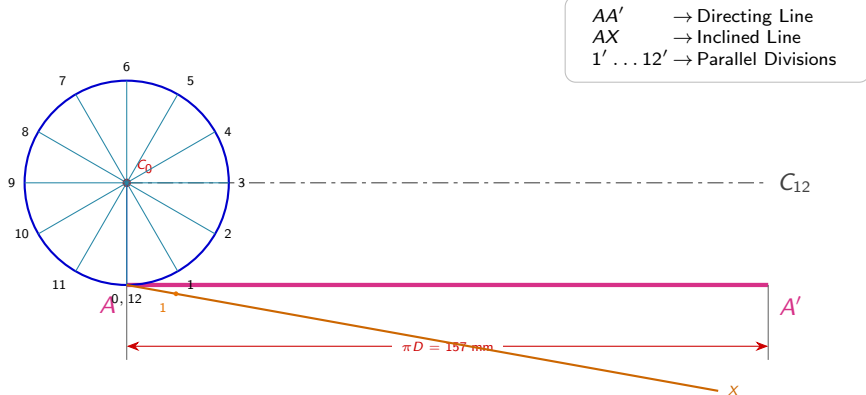
Step 3: Divide Directing Line using Line Division Method



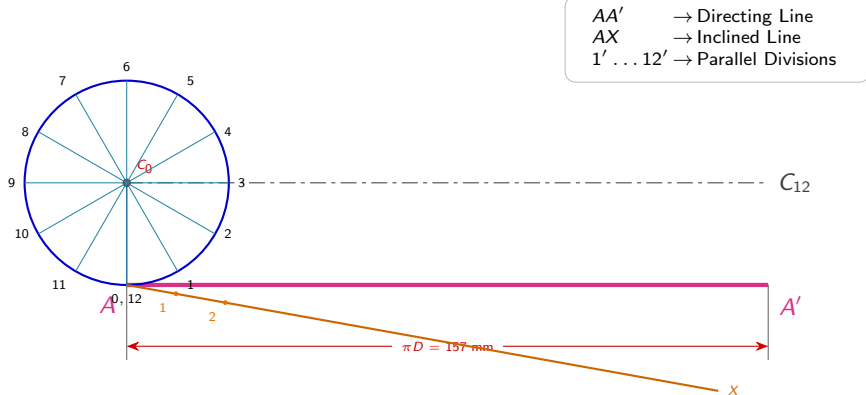
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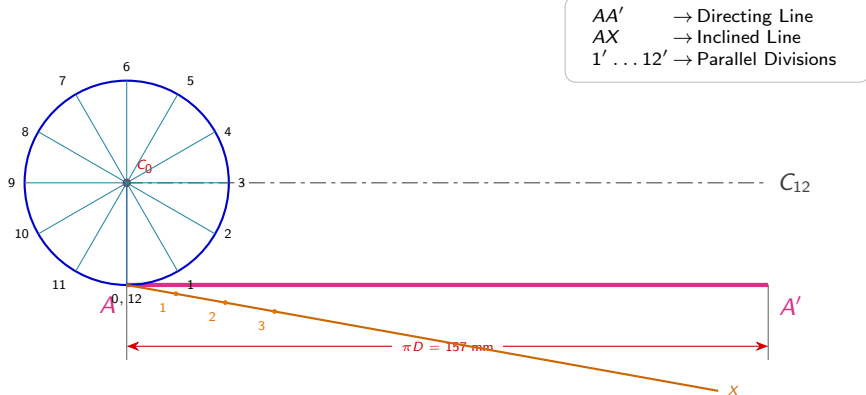
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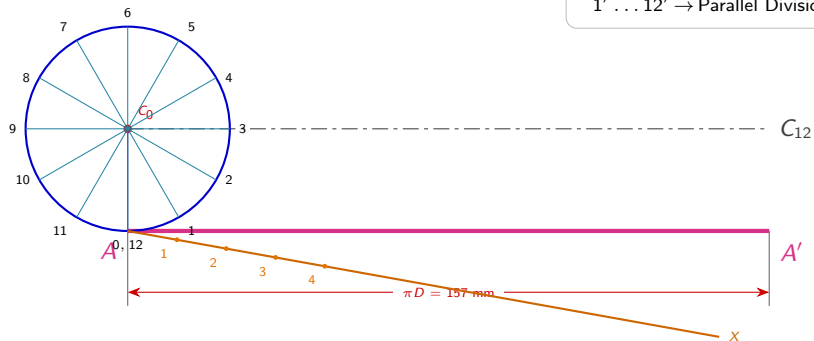
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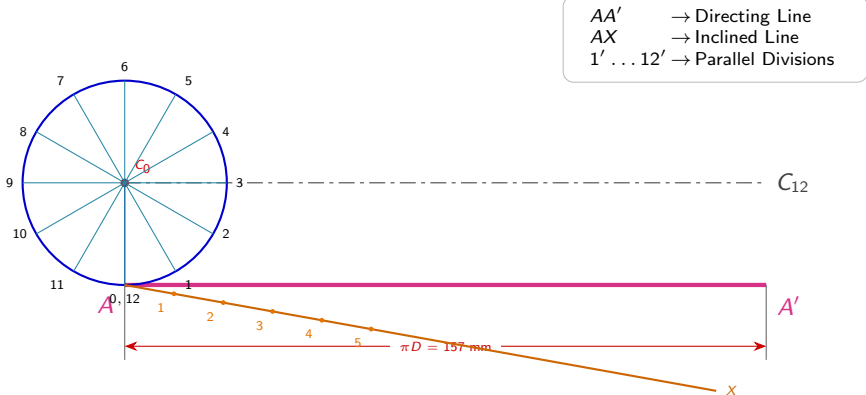
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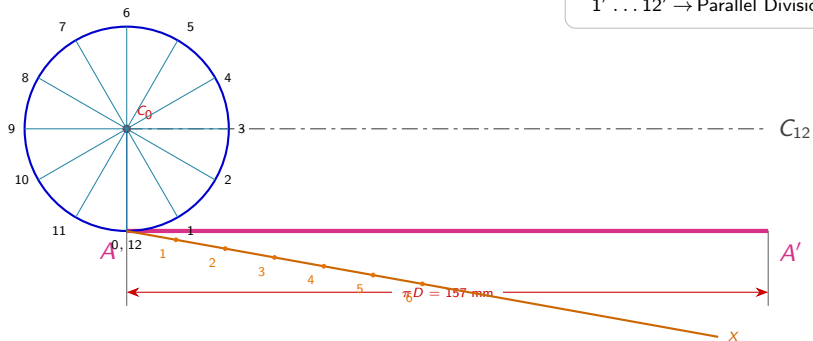
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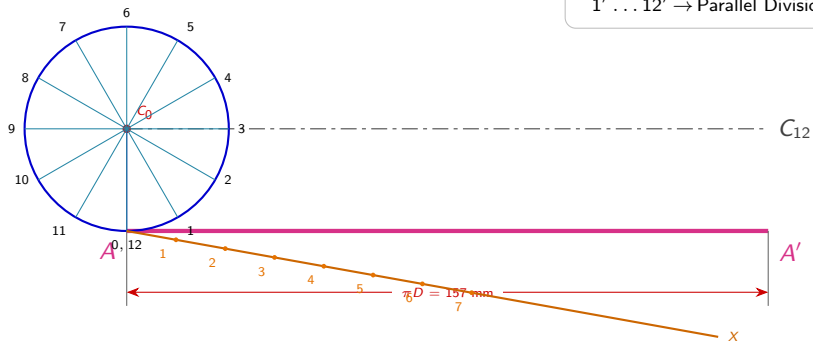
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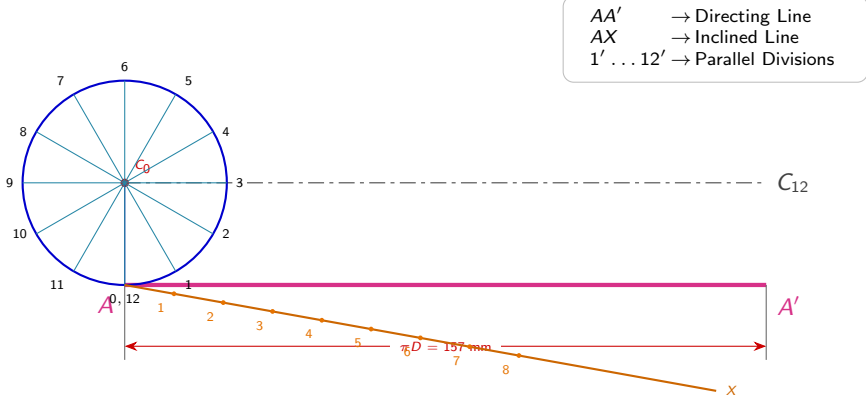
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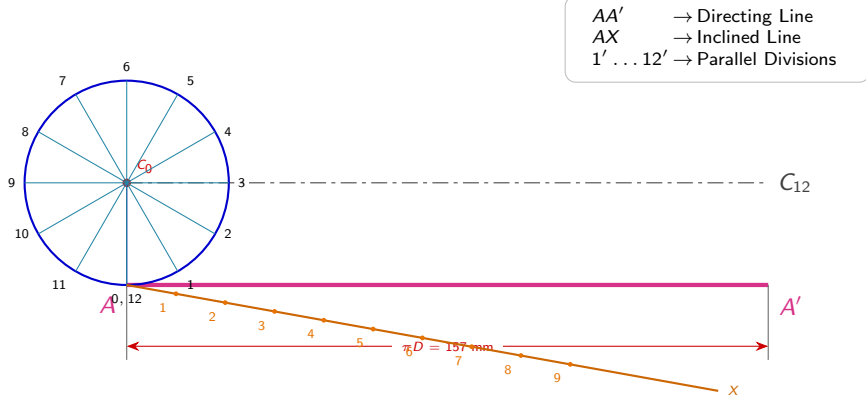
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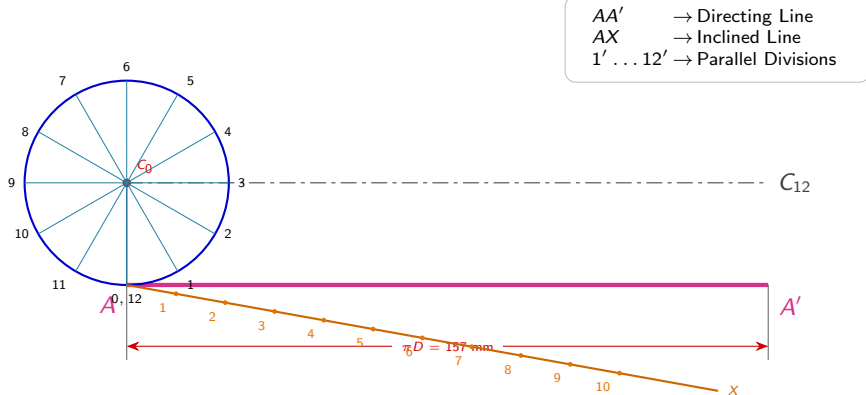
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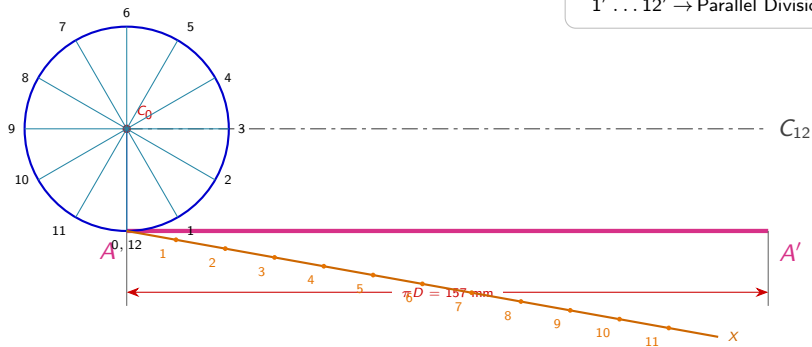
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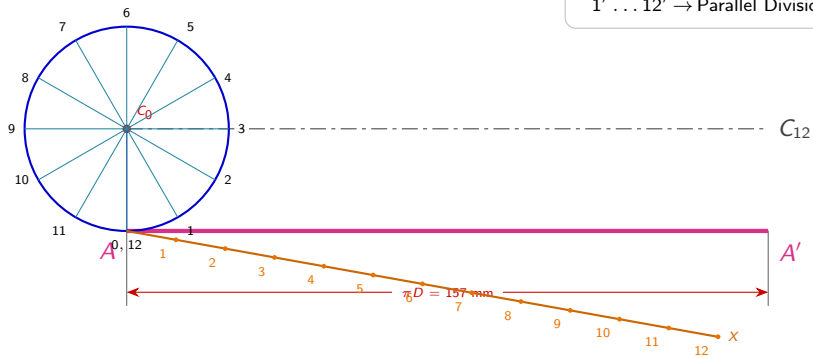
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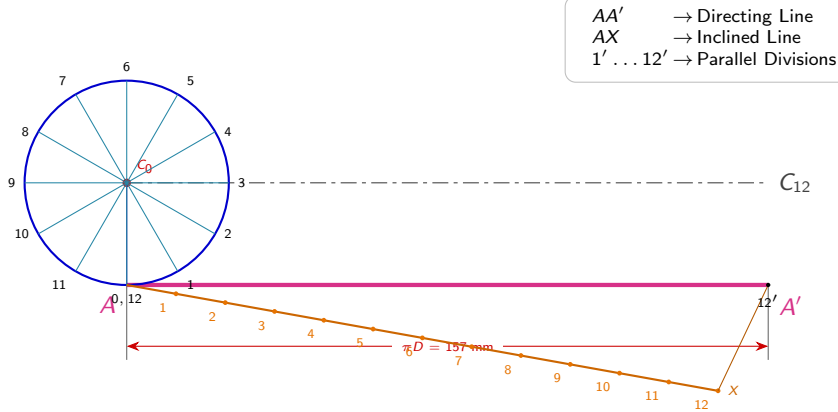
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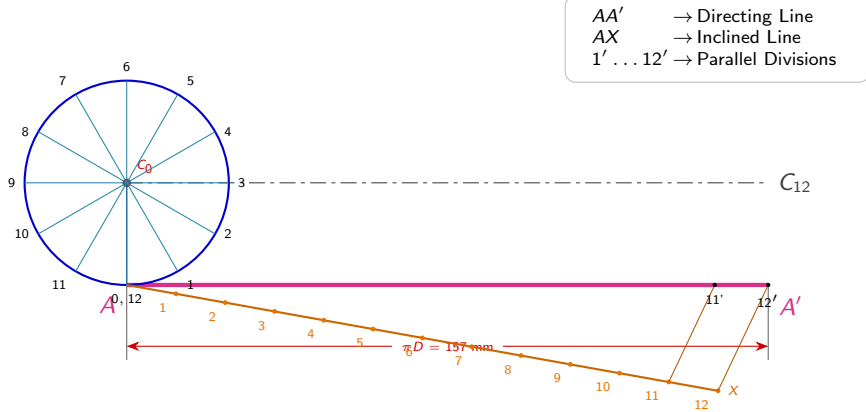
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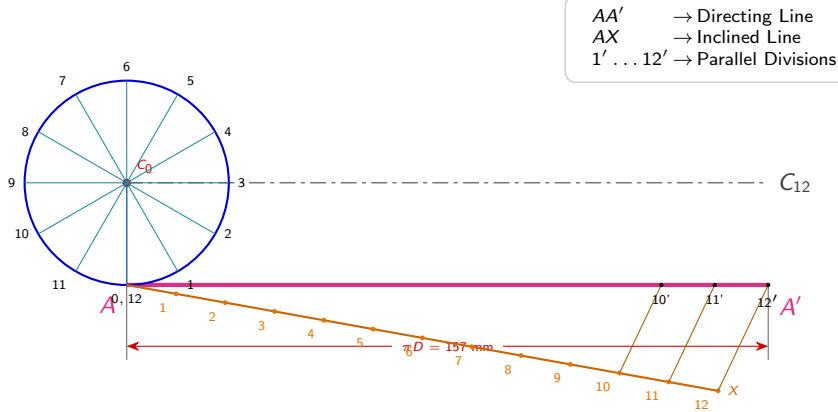
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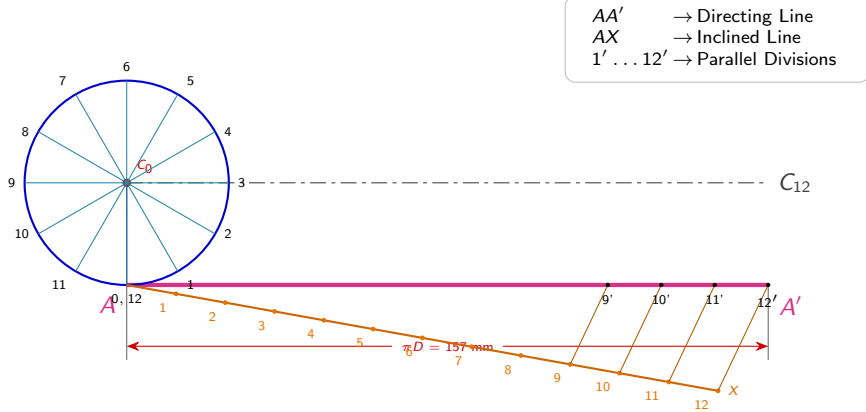
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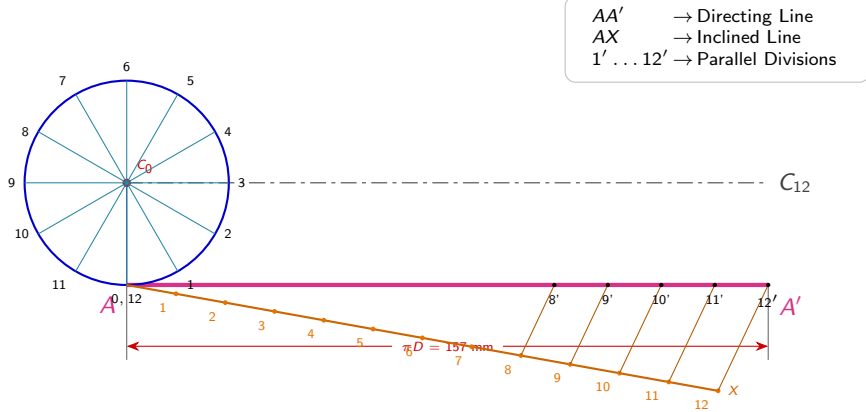
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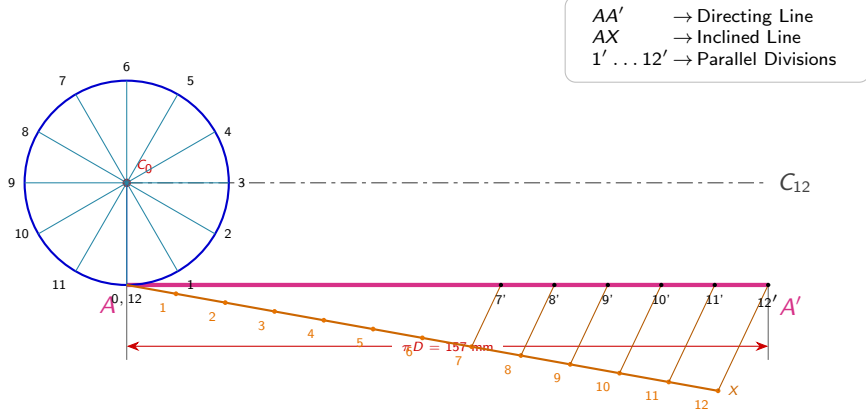
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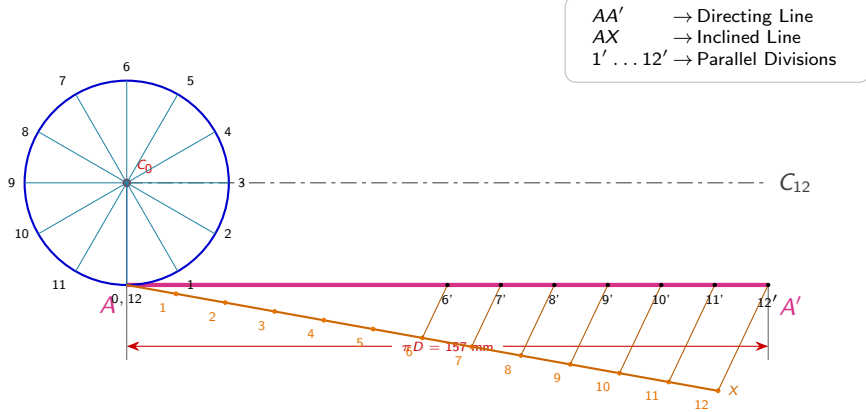
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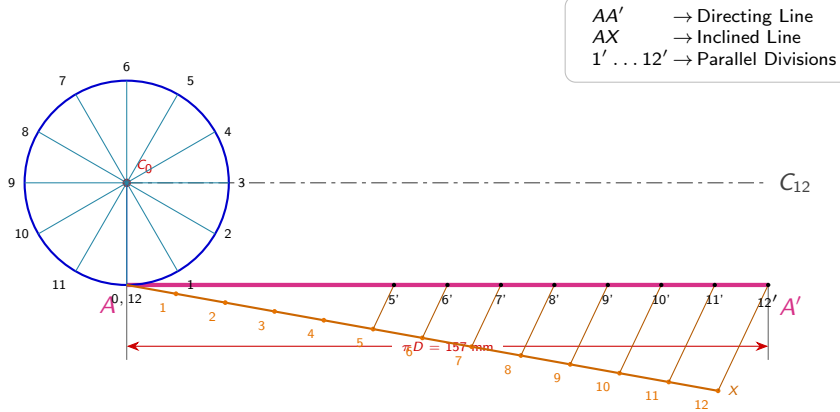
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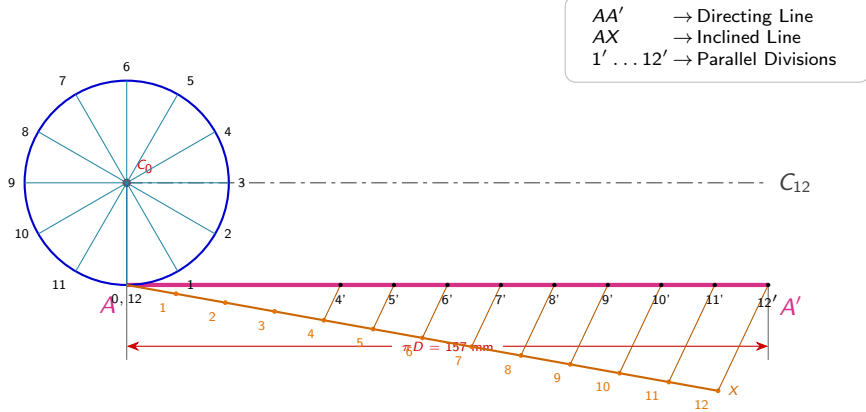
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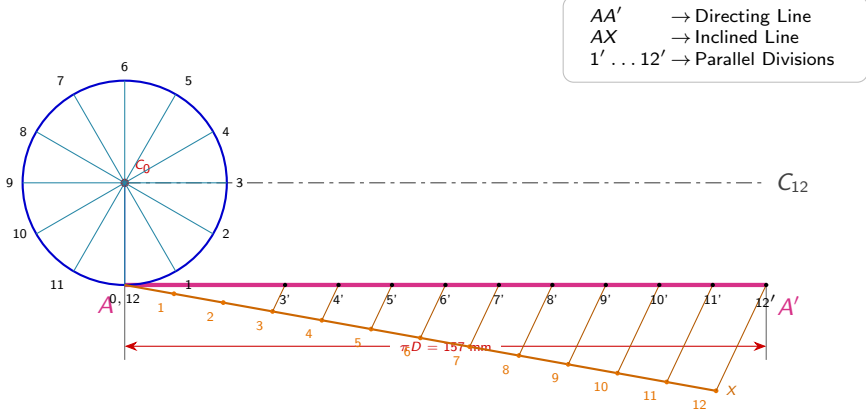
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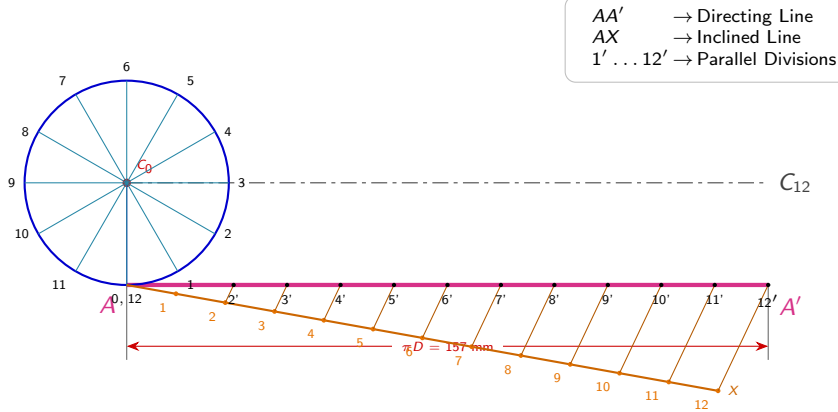
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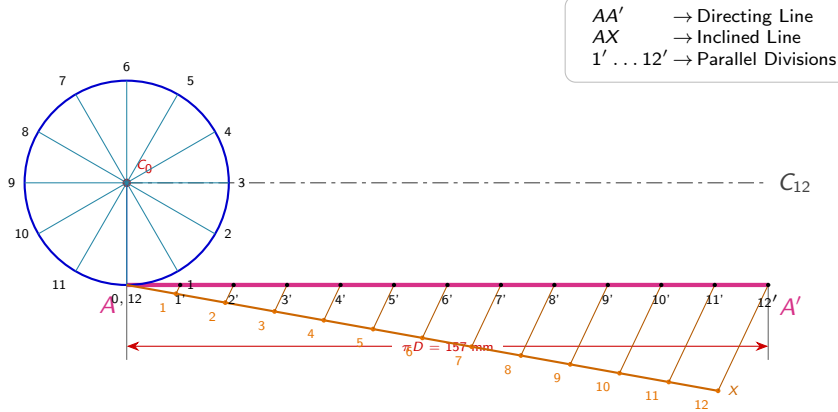
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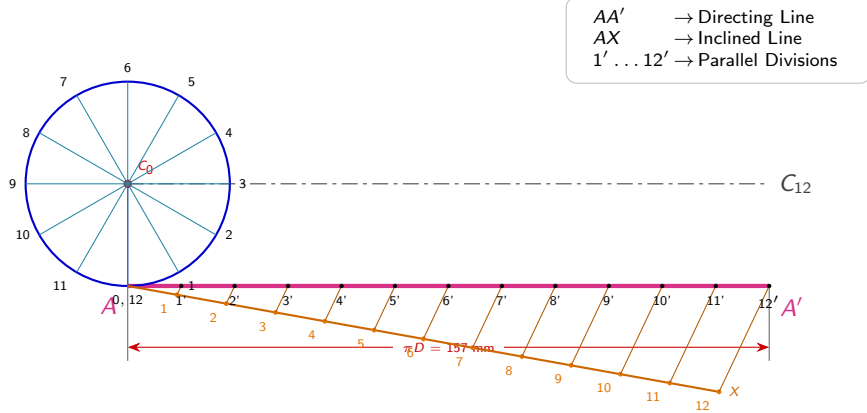
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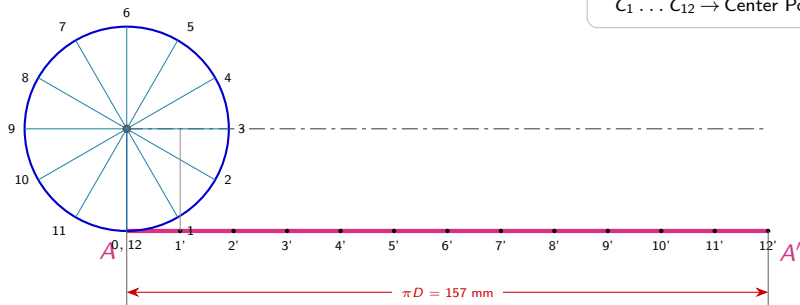
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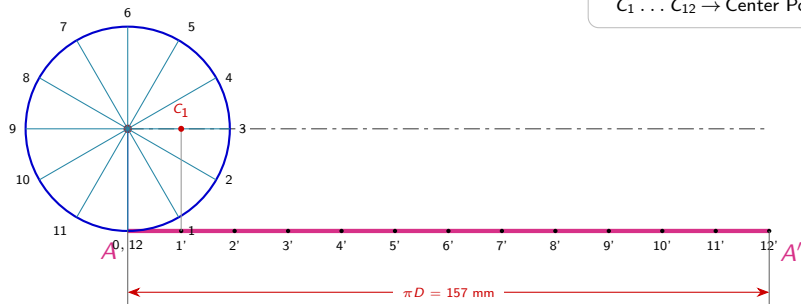
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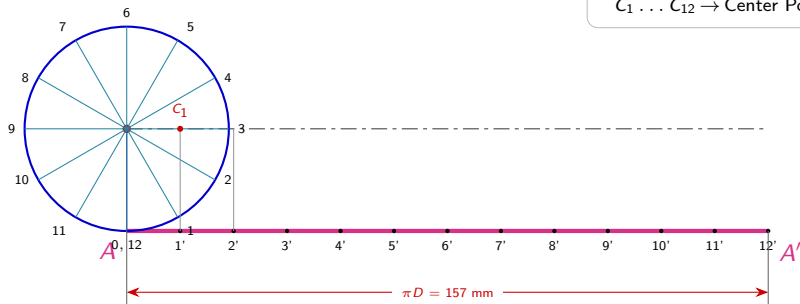
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



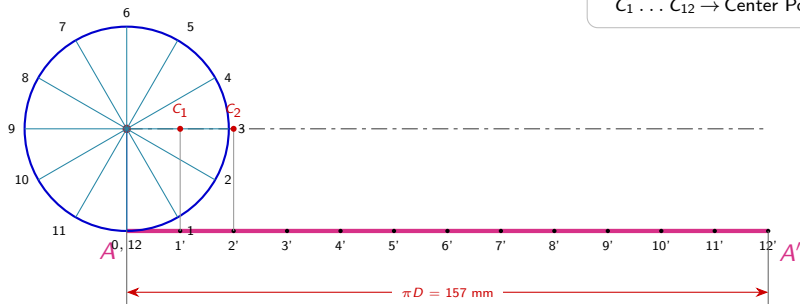
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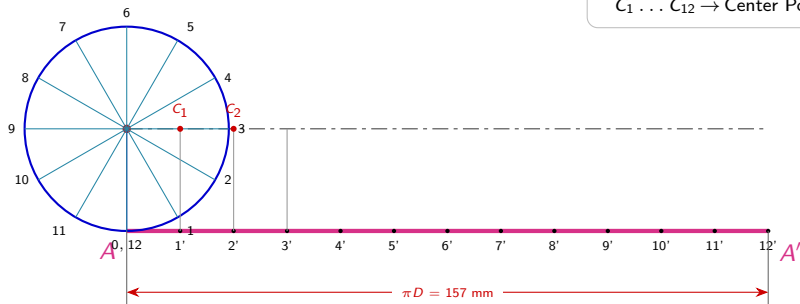
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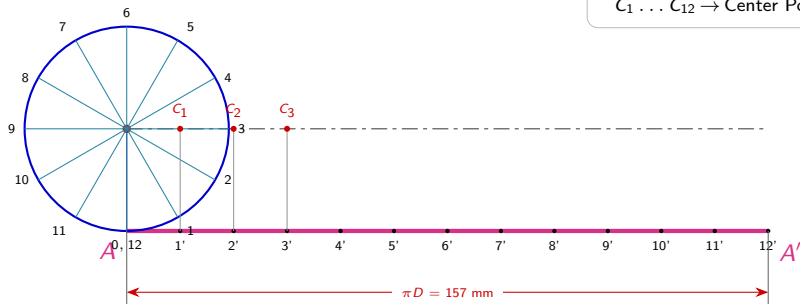
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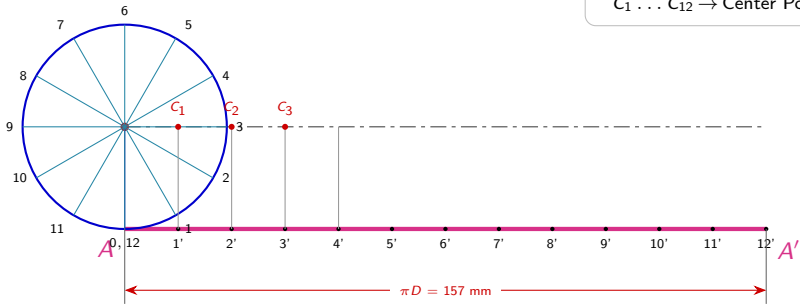
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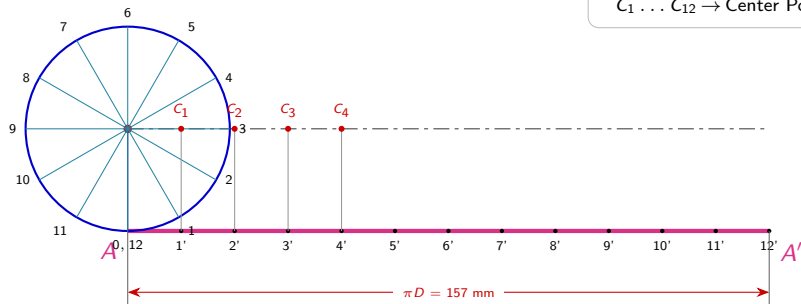
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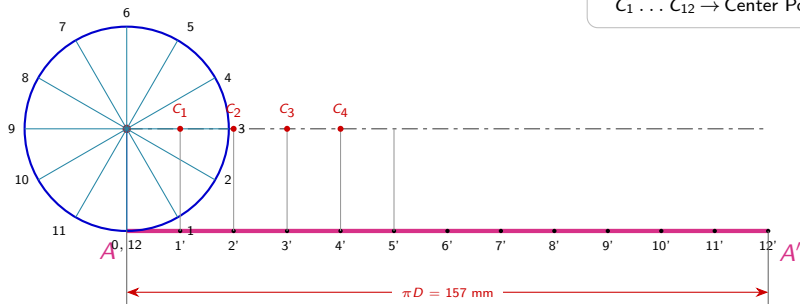
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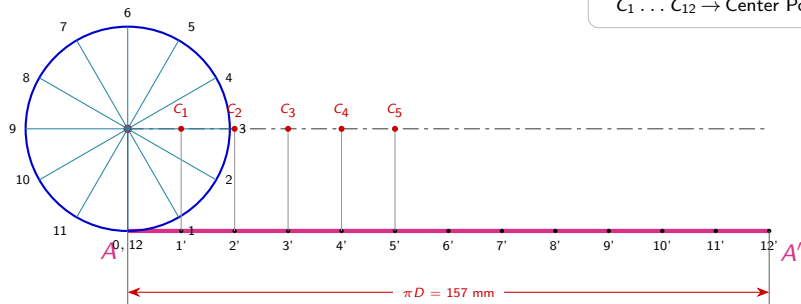
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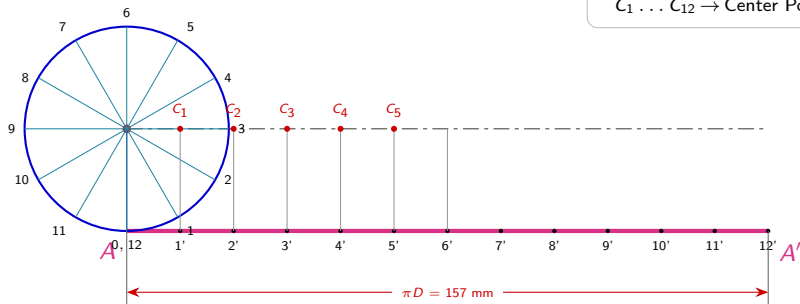
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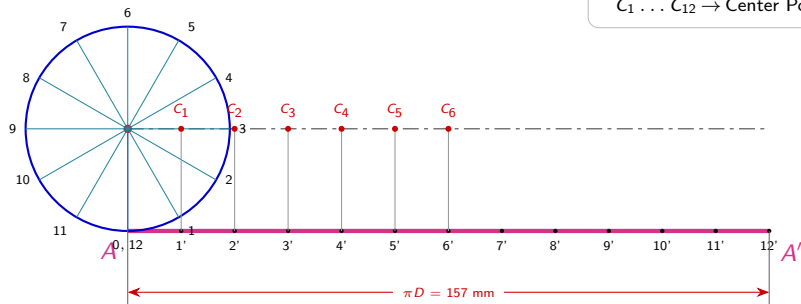
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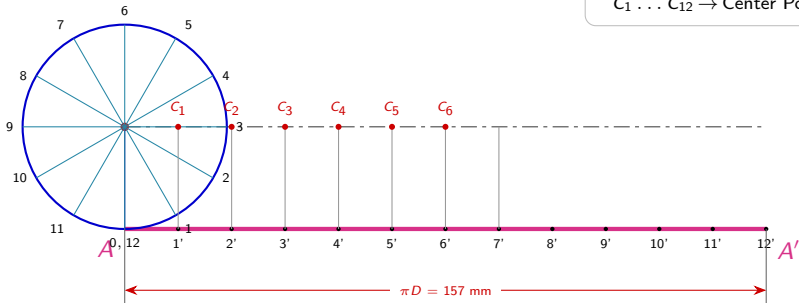
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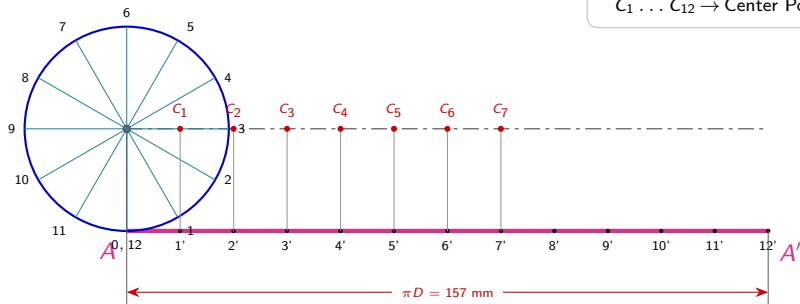


AA' → Directing Line

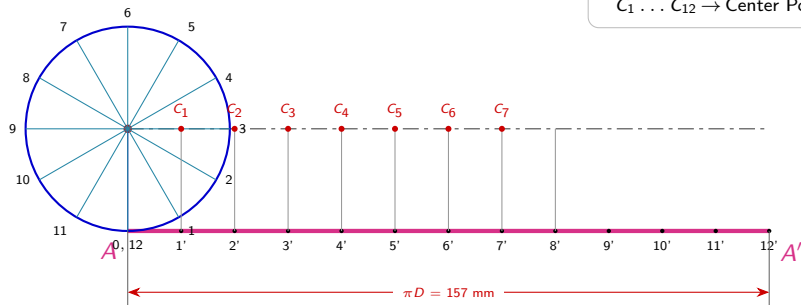
Axis Line → Center Line

 $C_1 \dots C_{12} \rightarrow \text{Center Points}$

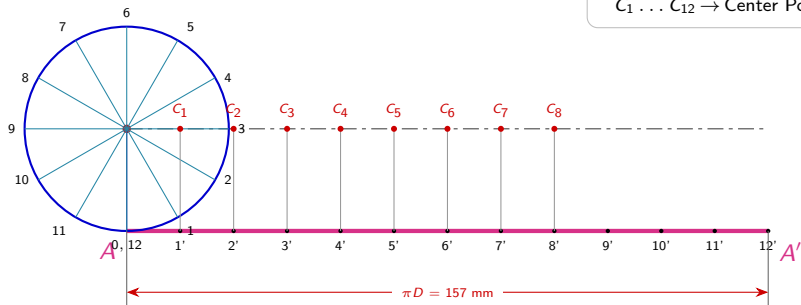
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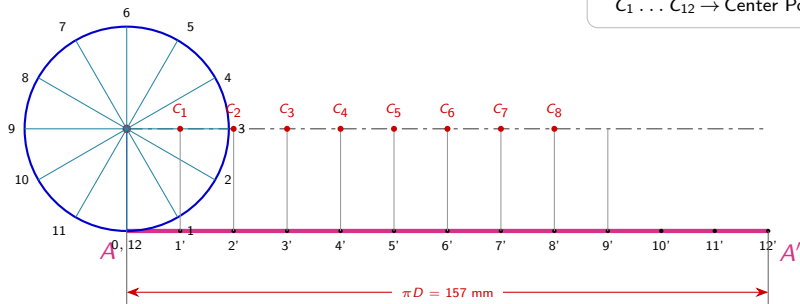
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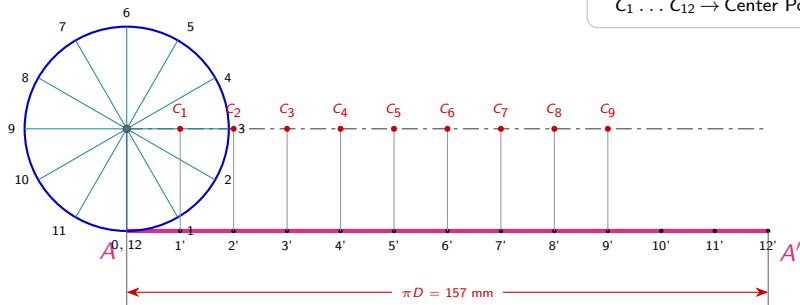
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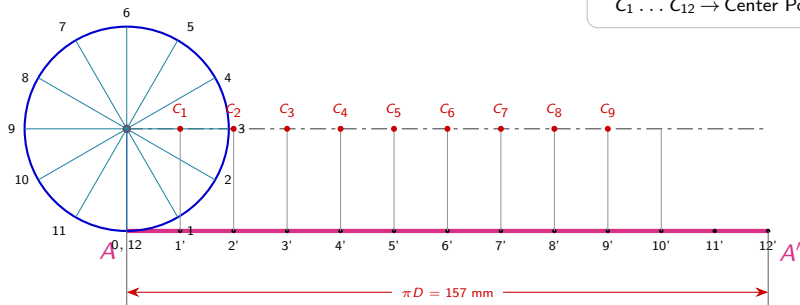
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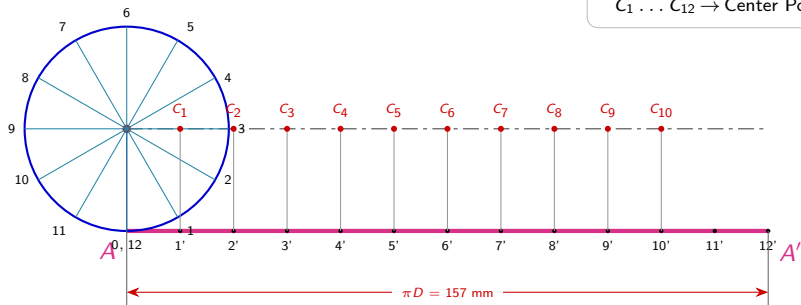
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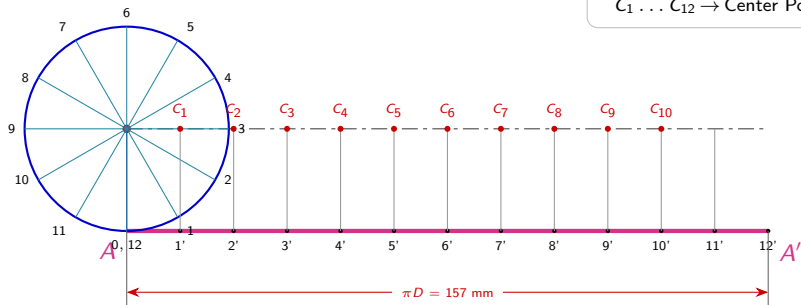
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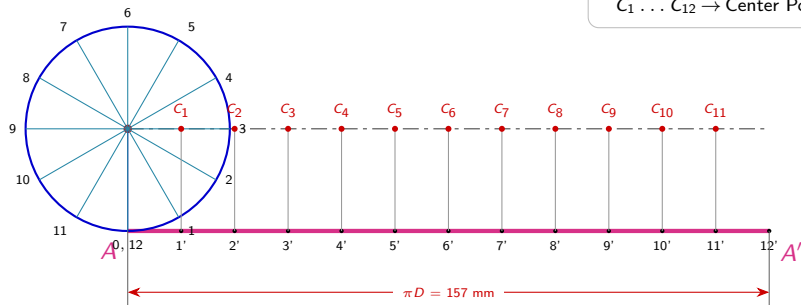
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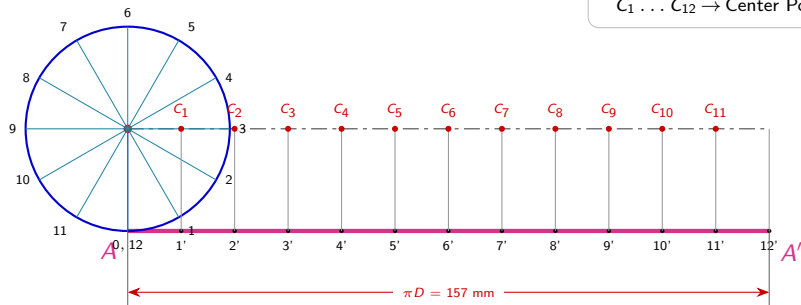
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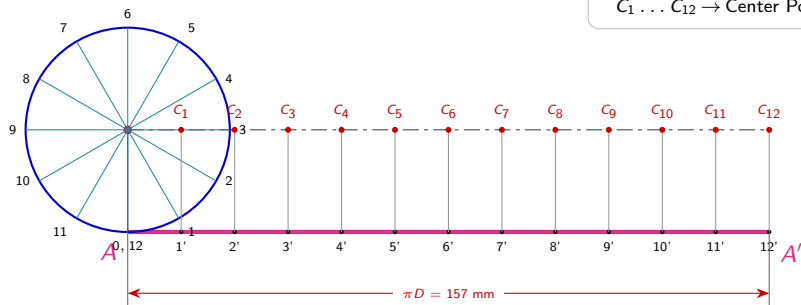
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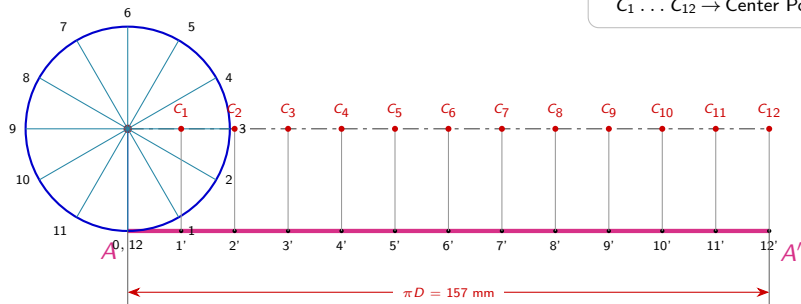
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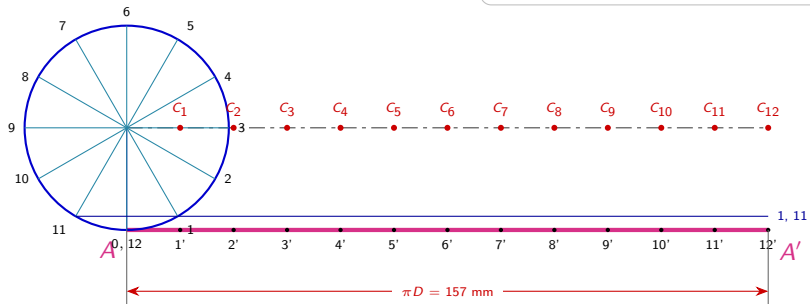


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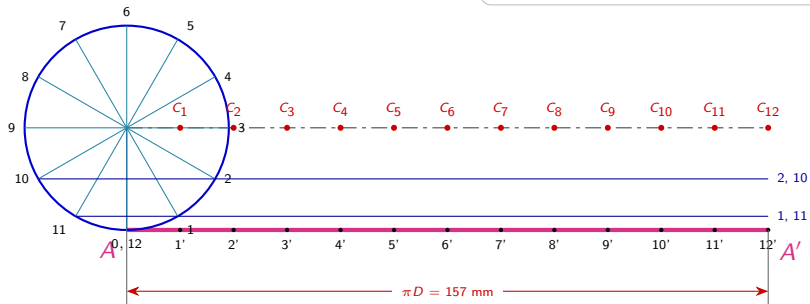
Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers



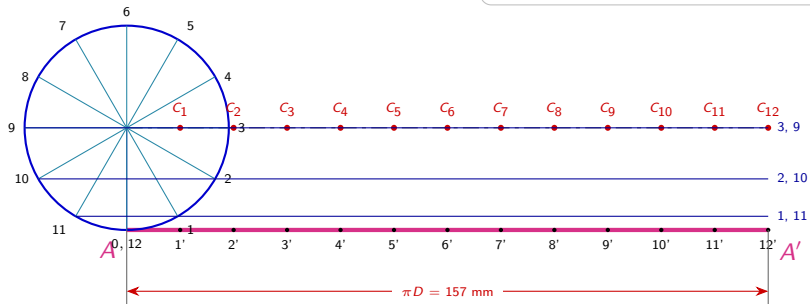
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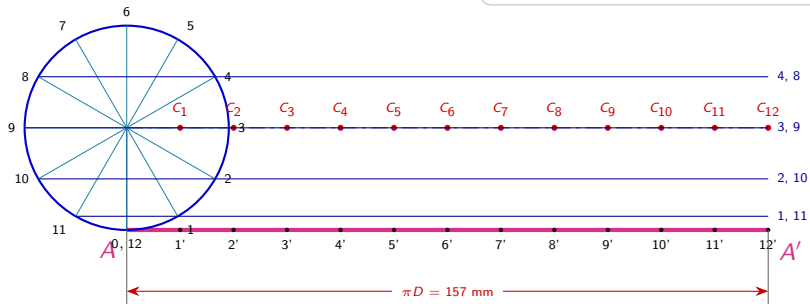
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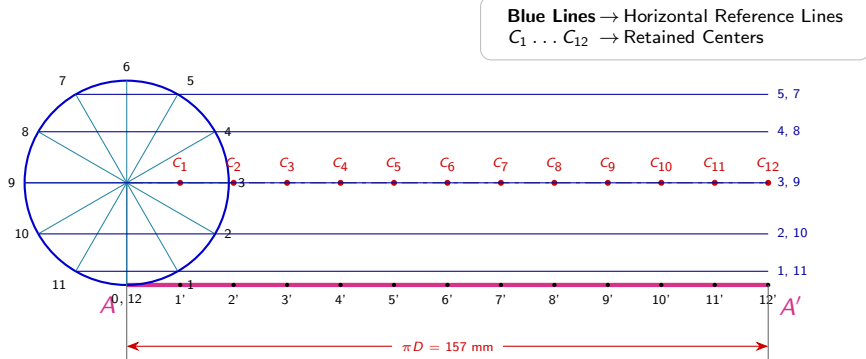


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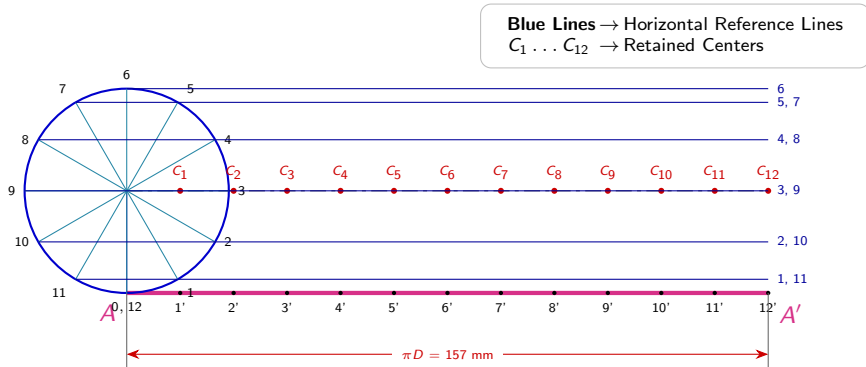
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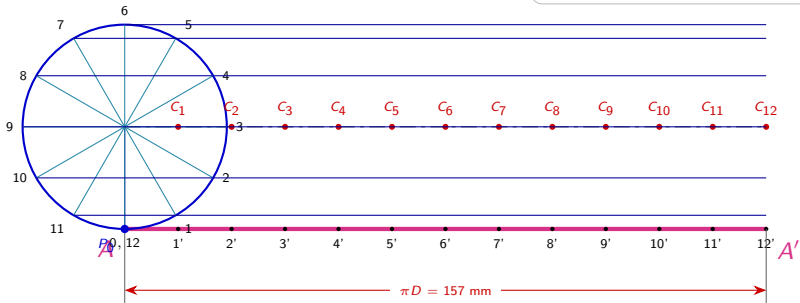
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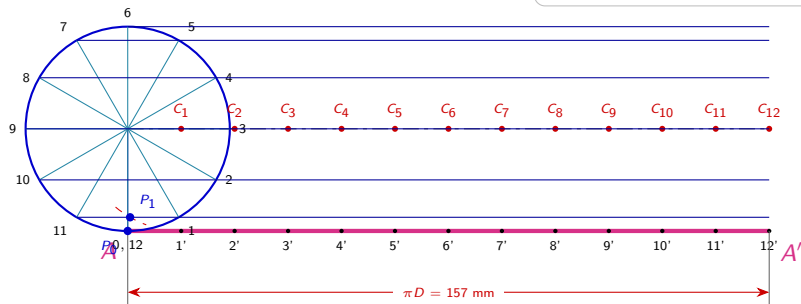
Step 5: Draw Horizontal Reference Lines



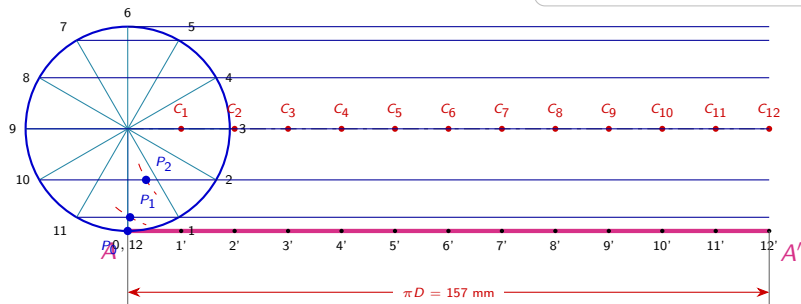
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$$\begin{aligned} C_1 \dots C_{12} &\rightarrow \text{Retained Centers} \\ P_1 \dots P_{12} &\rightarrow \text{Cutting Arcs/Points} \end{aligned}$$


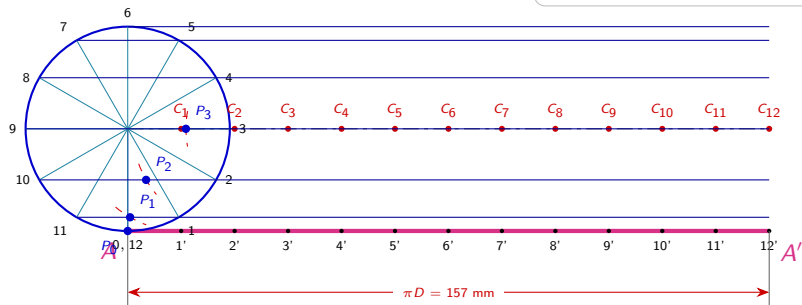
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



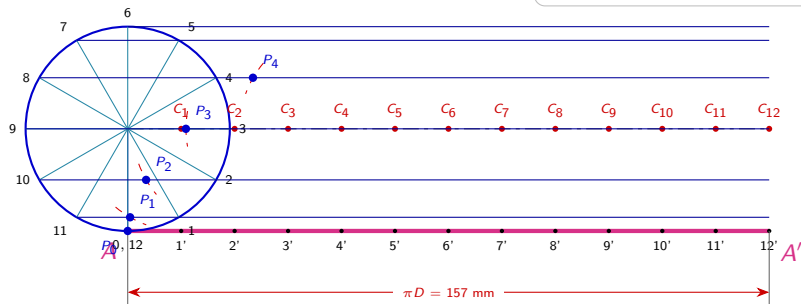
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



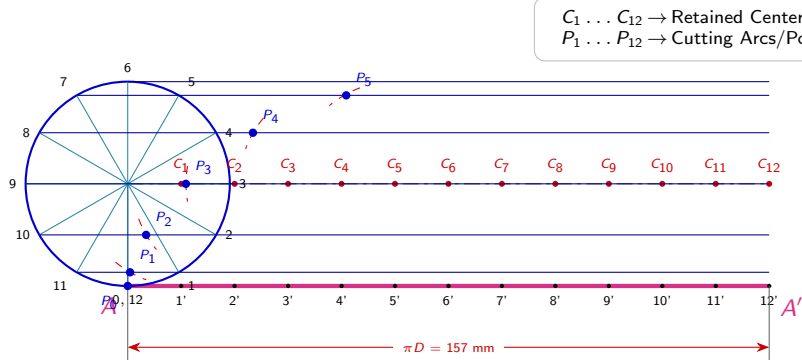
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



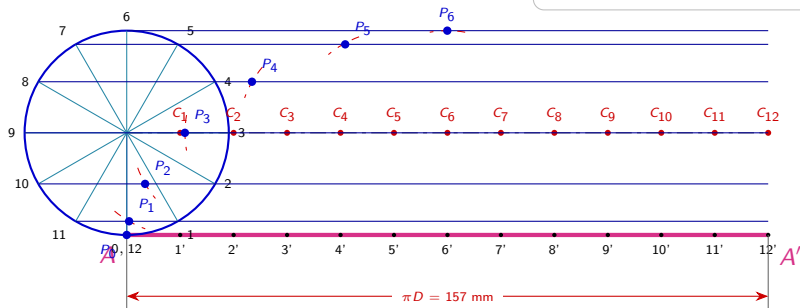
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



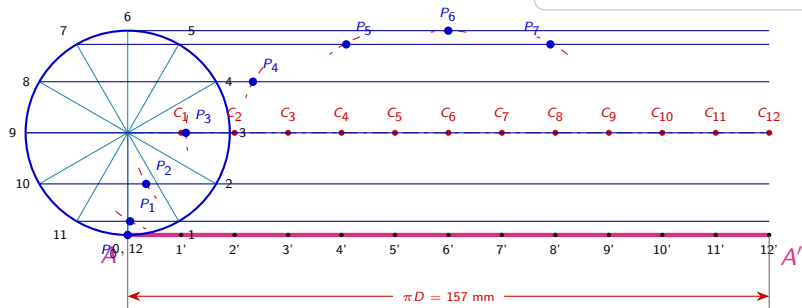
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



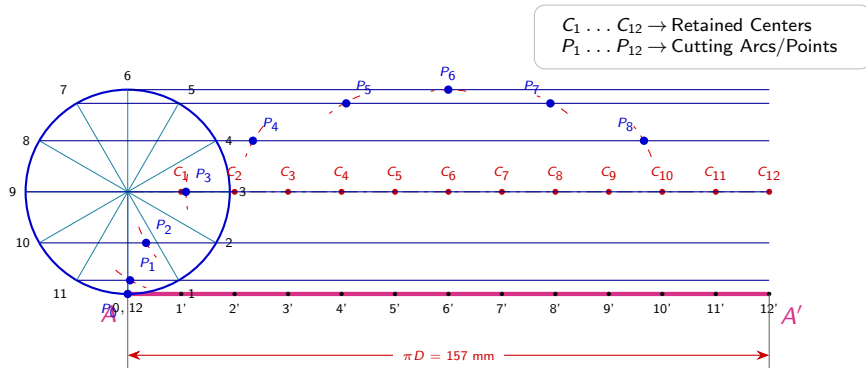
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



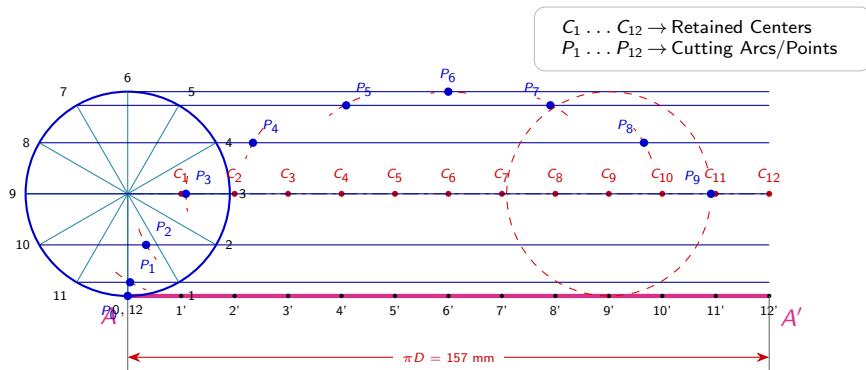
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



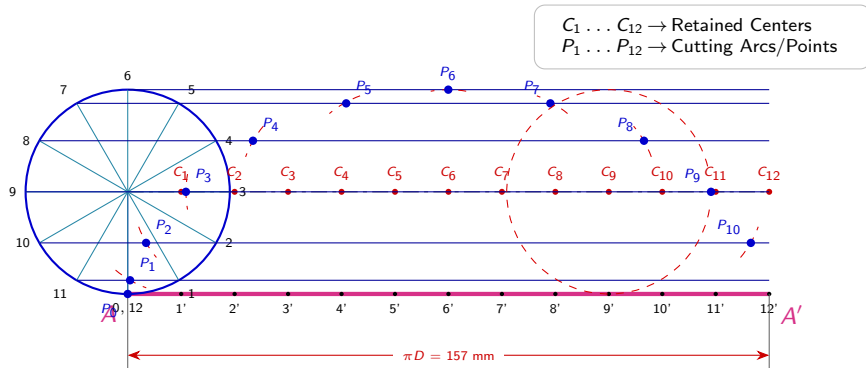
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



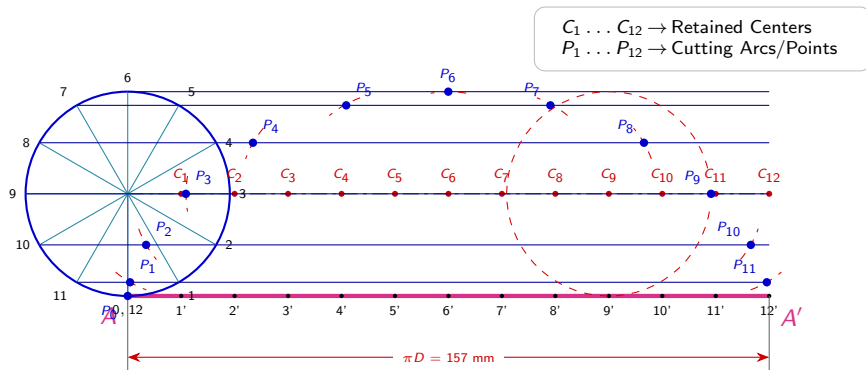
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



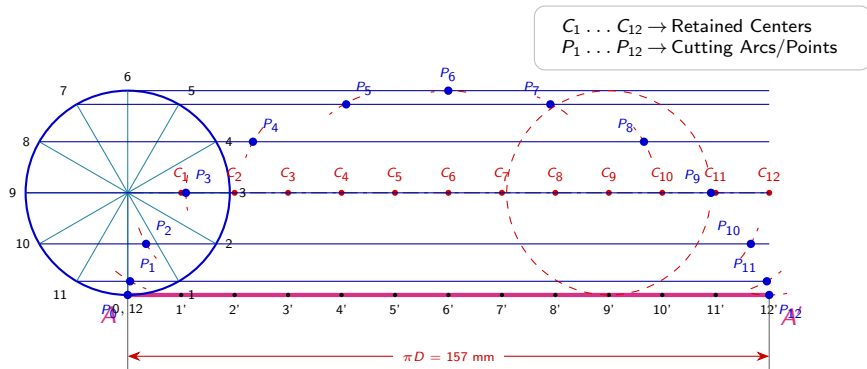
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



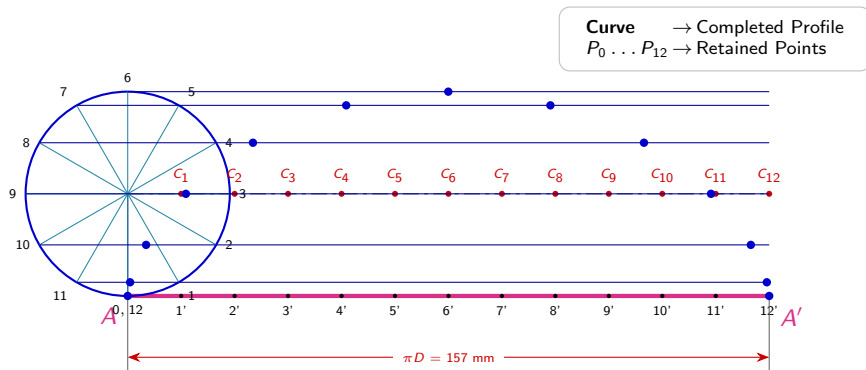
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



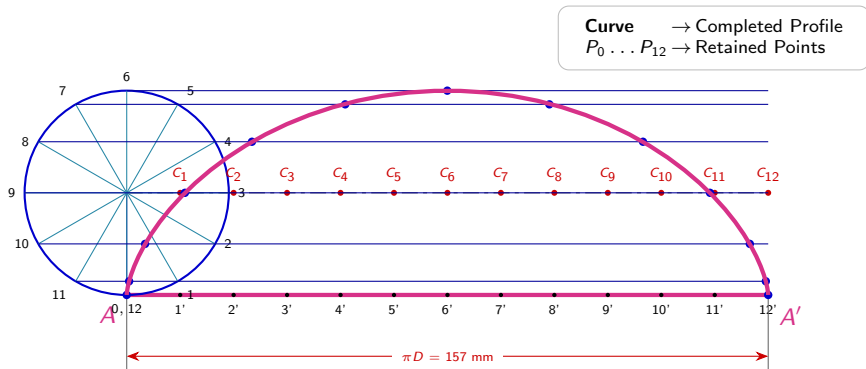
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



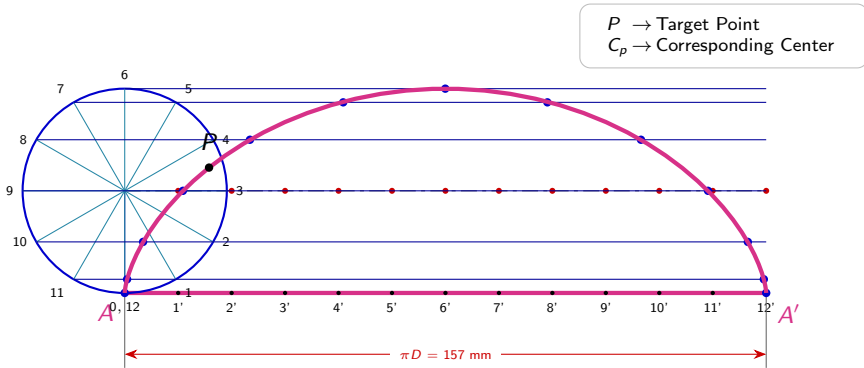
Step 7: Smoothly Join Points to Form Cycloid Curve



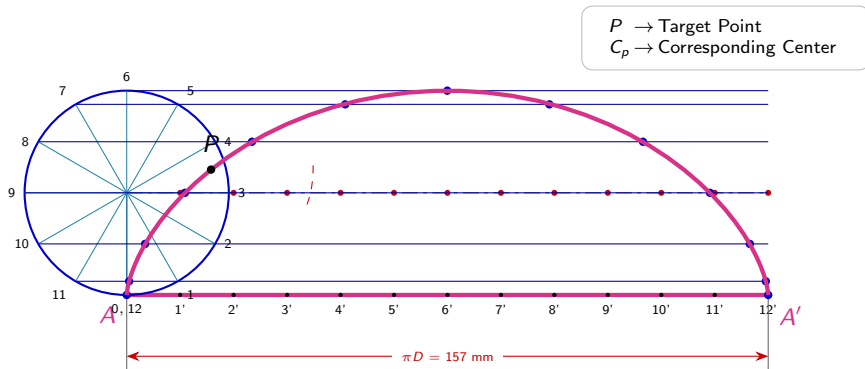
Step 7: Smoothly Join Points to Form Cycloid Curve



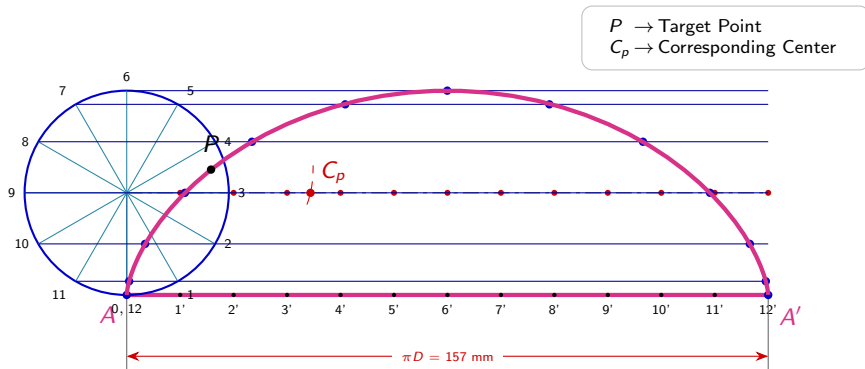
Step 8: Mark Point P & Locate Center C_p



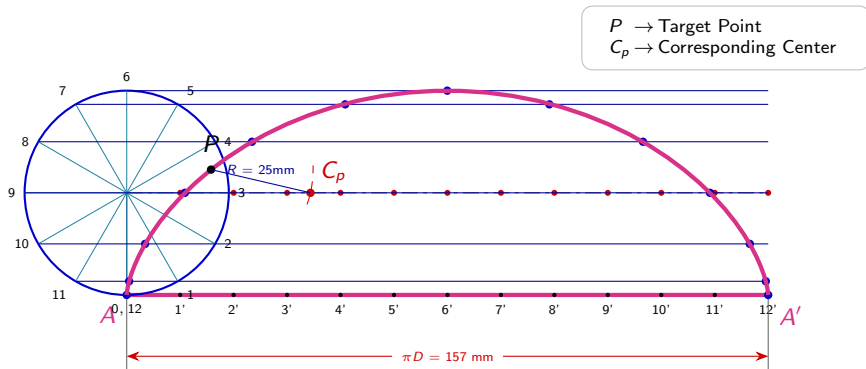
Step 8: Mark Point P & Locate Center C_p



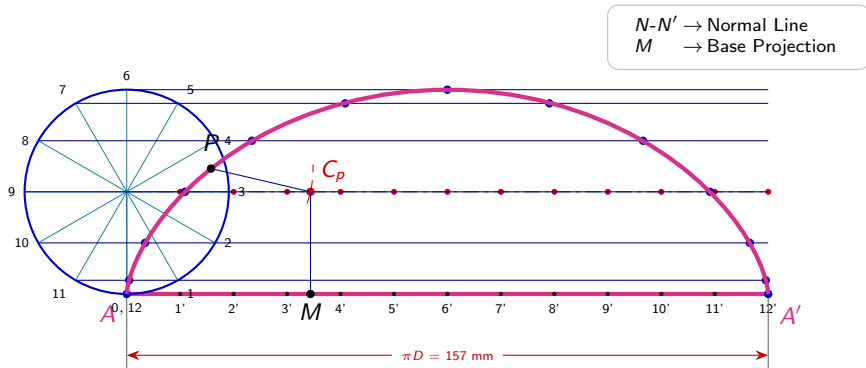
Step 8: Mark Point P & Locate Center C_p



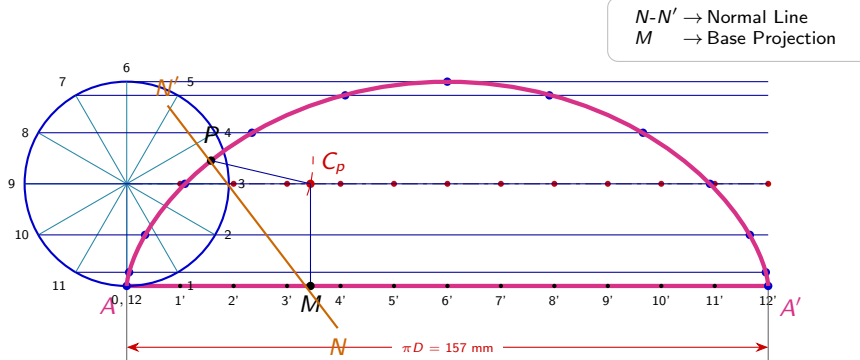
Step 8: Mark Point P & Locate Center C_p



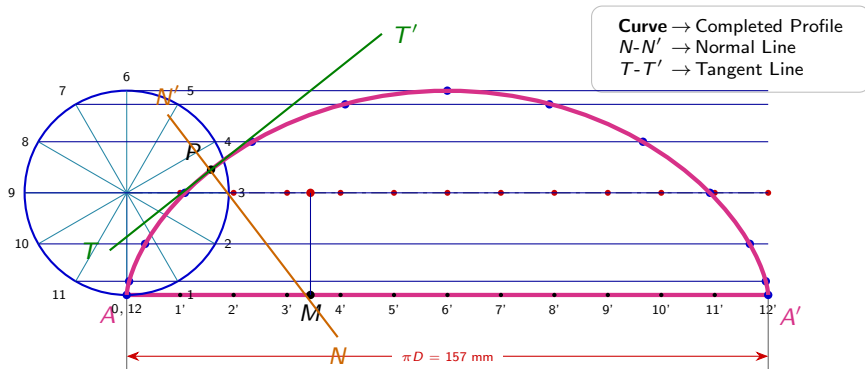
Step 9: Construct Normal Line $N-N'$



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Step 10: Construct Tangent Line $T-T'$ & Final Result



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